Week 10.1 | Reading the evidence

Collision-map errors and the low-temperature sampling domain

This executable lesson learns a Lennard–Jones deflection map; it does not run a learned collision kernel inside a DSMC flow simulation. Impact parameter b and relative energy E determine deflection angle χ. The surrogate predicts $\cos\chi$, so inspect its largest local errors near orbiting and glory features, not only a thermal average.

An optimizer stopping at its iteration budget is not a convergence certificate. Record the warning, iteration count, seed and library versions. Reproducing a retained decimal is not a reason to tune on final energies.

The transport audit samples a Maxwellian relative-energy distribution and drops states outside the tabulated interval. Report the number and fraction dropped at every temperature, separately below and above the table. At low temperature, a larger omitted low-energy mass can make a truncated integral look artificially accurate when reference and surrogate omit the same region. Compare the two on the same stated interval, then assess the omitted contribution separately. Diffusion and viscosity weight scattering differently; a single small integrated error cannot bound both cross sections at every energy.

Check your understanding. Identify the input information, the reference, the metric denominator and the evaluation split in one worked notebook comparison. Predict which conclusion would change if any one of those four changed, then check your prediction.